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November 11, 2025, 2:32 PM

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CAS Lexicon Search:

Benzylic oxidation - Preferred Concept

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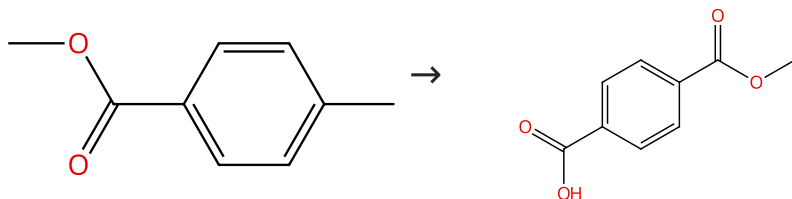
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 99%


 Suppliers (89)

 Suppliers (88)

31-488-CAS-4494966

Steps: 1 Yield: 99%

Direct and Selective Benzylic Oxidation of Alkylarenes via C-H Abstraction Using Alkali Metal Bromides

By: Moriyama, Katsuhiko; et al

Organic Letters (2012), 14(9), 2414-2417.

1.1 **Reagents:** Potassium bromide, Potassium peroxymonosulfate sulfate (2KHSO₅.KHSO₄.K₂SO₄)

Solvents: Dichloromethane, Water; 24 h, rt

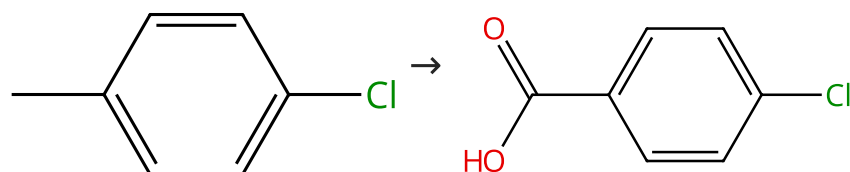
1.2 **Reagents:** Sodium sulfite

Solvents: Water; rt

Experimental Protocols

Scheme 2 (2 Reactions)

Steps: 1 Yield: 95-99%


 Suppliers (82)

 Suppliers (105)

31-488-CAS-8756059

Steps: 1 Yield: 99%

Direct and Selective Benzylic Oxidation of Alkylarenes via C-H Abstraction Using Alkali Metal Bromides

By: Moriyama, Katsuhiko; et al

Organic Letters (2012), 14(9), 2414-2417.

1.1 **Reagents:** Potassium bromide, Potassium peroxymonosulfate sulfate (2KHSO₅.KHSO₄.K₂SO₄)

Solvents: Dichloromethane, Water; 24 h, rt

1.2 **Reagents:** Sodium sulfite

Solvents: Water; rt

Experimental Protocols

31-614-CAS-35375709

Steps: 1 Yield: 95%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

By: Wu, Jieqing; et al

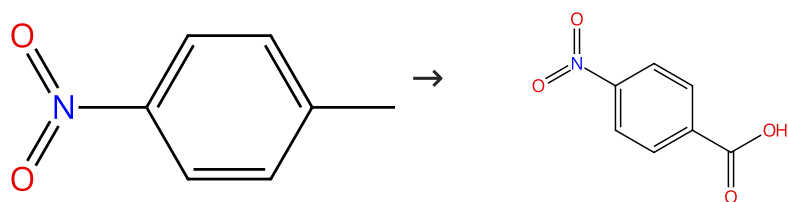
Green Chemistry (2023), 25(3), 940-945.

1.1 **Reagents:** Oxygen
Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)
Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (75)

Suppliers (105)

31-488-CAS-11469892

Steps: 1 Yield: 99%

Direct and Selective Benzylic Oxidation of Alkylarenes via C-H Abstraction Using Alkali Metal Bromides

By: Moriyama, Katsuhiko; et al

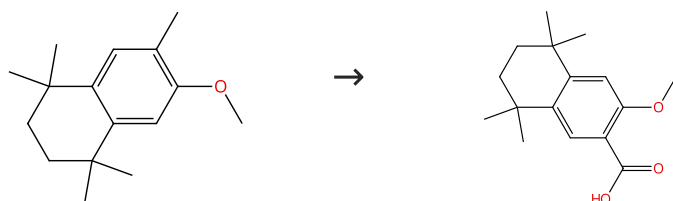
Organic Letters (2012), 14(9), 2414-2417.

1.1 **Reagents:** Potassium bromide, Potassium peroxymonosulfate sulfate (2KHSO₅.KHSO₄.K₂SO₄)**Solvents:** Dichloromethane, Water; 24 h, rt1.2 **Reagents:** Sodium sulfite**Solvents:** Water; rt

Experimental Protocols

Scheme 4 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (8)

Suppliers (3)

31-488-CAS-955771

Steps: 1 Yield: 95%

Concise synthesis of the retinoic acid receptor (RAR) agonist A GN-193836 utilizing a photochemical benzylic oxidation

By: Fish, Paul V.; et al

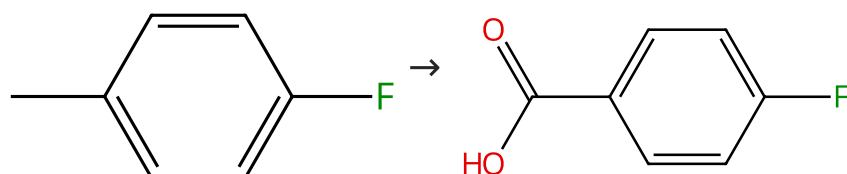
Green Chemistry Letters and Reviews (2012), 5(4), 535-538.

1.1 **Reagents:** Oxygen**Catalysts:** Allyl bromide**Solvents:** Ethyl acetate; 48 h

Experimental Protocols

Scheme 5 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (62)

Suppliers (97)

31-614-CAS-35375711

Steps: 1 Yield: 94%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

By: Wu, Jieqing; et al

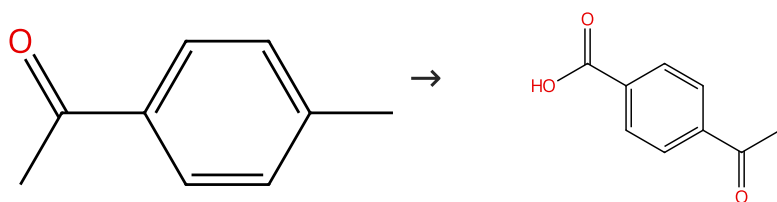
Green Chemistry (2023), 25(3), 940-945.

1.1 **Reagents:** Oxygen**Catalysts:** Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)**Solvents:** Acetonitrile; 6 h, 1 atm, 20 °C

Experimental Protocols

Scheme 6 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (103)

Suppliers (101)

31-614-CAS-35375719

Steps: 1 Yield: 94%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

By: Wu, Jieqing; et al

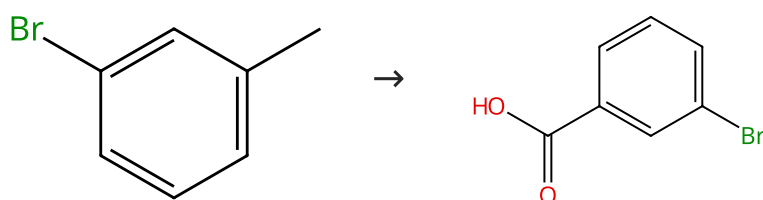
Green Chemistry (2023), 25(3), 940-945.

1.1 **Reagents:** Oxygen
Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)
Solvents: Acetonitrile; 18 h, 1 atm, 20 °C

Experimental Protocols

Scheme 7 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (58)

Suppliers (95)

31-488-CAS-9034326

Steps: 1 Yield: 93%

Direct and Selective Benzylic Oxidation of Alkylarenes via C-H Abstraction Using Alkali Metal Bromides

By: Moriyama, Katsuhiko; et al

Organic Letters (2012), 14(9), 2414-2417.

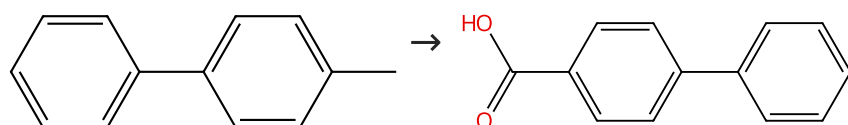
1.1 **Reagents:** Potassium bromide, Potassium peroxymonosulfate sulfate (2KHSO₅.KHSO₄.K₂SO₄)
Solvents: Dichloromethane, Water; 24 h, rt

1.2 **Reagents:** Sodium sulfite
Solvents: Water; rt

Experimental Protocols

Scheme 8 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (82)

Suppliers (101)

31-614-CAS-35375708

Steps: 1 Yield: 93%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

By: Wu, Jieqing; et al

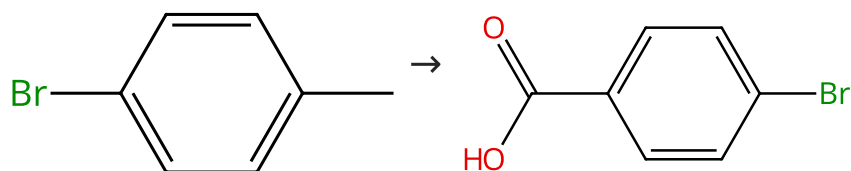
Green Chemistry (2023), 25(3), 940-945.

1.1 **Reagents:** Oxygen
Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)
Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

Experimental Protocols

Scheme 9 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (67)

Suppliers (98)

31-614-CAS-35375713

Steps: 1 Yield: 93%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

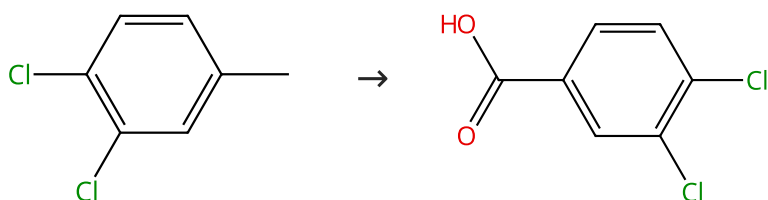
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 10 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (56)

Suppliers (81)

31-614-CAS-35375717

Steps: 1 Yield: 92%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

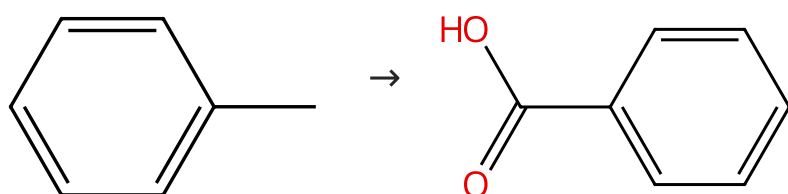
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (280)

Suppliers (175)

31-614-CAS-35375706

Steps: 1 Yield: 91%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

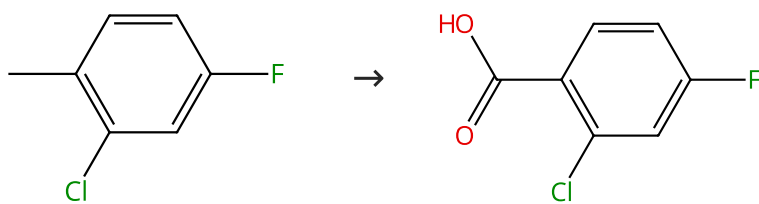
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (78)

Suppliers (82)

31-614-CAS-35375716

Steps: 1 Yield: 90%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

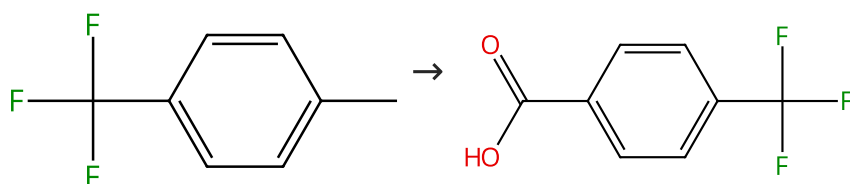
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (69)

Suppliers (91)

31-614-CAS-35375718

Steps: 1 Yield: 90%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 18 h, 1 atm, 20 °C

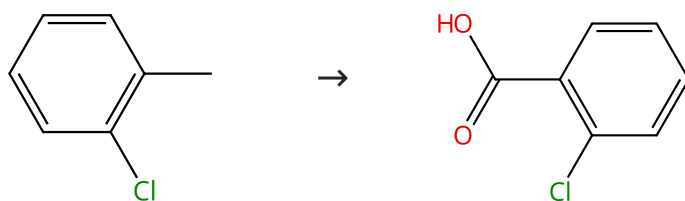
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (71)

Suppliers (97)

31-614-CAS-35375712

Steps: 1 Yield: 90%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

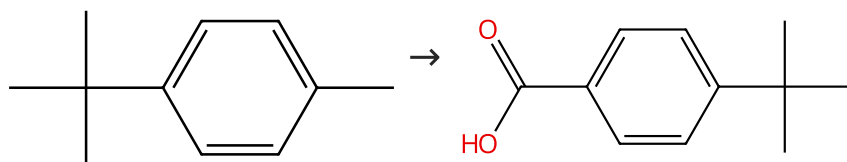
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 89%



Suppliers (63)

Suppliers (92)

31-614-CAS-35375707

Steps: 1 Yield: 89%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

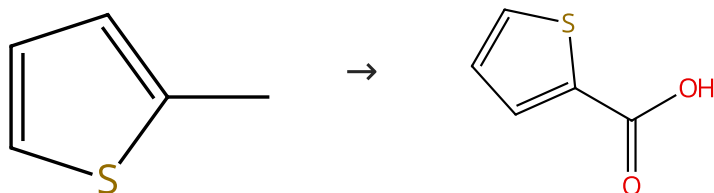
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 88%



Suppliers (70)

Suppliers (95)

31-614-CAS-35375728

Steps: 1 Yield: 88%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

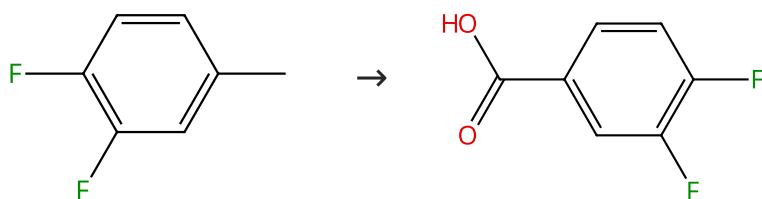
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 87%



Suppliers (75)

Suppliers (90)

31-614-CAS-35375714

Steps: 1 Yield: 87%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

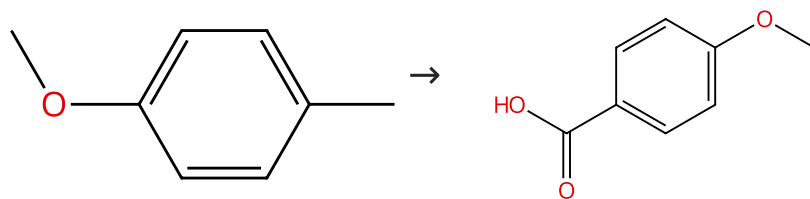
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 18 (6 Reactions)

Steps: 1 Yield: 53-87%



Suppliers (79)

Suppliers (109)

31-614-CAS-35375710

Steps: 1 Yield: 87%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

- 1.1 **Reagents:** Oxygen
Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)
Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

Experimental Protocols

31-614-CAS-35034782

Steps: 1 Yield: 80%

Chemoselective benzylic oxidation of alkyl-substituted aromatics with singlet molecular oxygen generated from trans-3,5 hydroperoxy-3,5 dimethyl-1,2-dioxolan-3-yl ethaneperoxate

By: Najminejad, Zohreh

Synthetic Communications (2022), 52(24), 2301-2310.

- 1.1 **Reagents:** Potassium hydroxide, *rel*-(3*R*,5*S*)-5-Hydroperoxy-3,5-dimethyl-1,2-dioxolan-3-yl ethaneperoxate
Solvents: Water, Palladium, bis(acetonitrile)dichloro-; 50 min, rt
- 1.2 **Reagents:** Sodium sulfite
Solvents: Water; rt; 10 min, rt

Experimental Protocols

31-614-CAS-33766772

Steps: 1 Yield: 80%

Chemoselective Benzylic Oxidation of Alkyl-Substituted Aromatics to Ketones with trans-3,5-Dihydroperoxy-3,5 Dimethyl-1,2-Dioxolane

By: Najminejad, Z.; et al

Organic Preparations and Procedures International (2022), 54(6), 511-516.

- 1.1 **Reagents:** *rel*-(3*R*,5*R*)-3,5-Bis(hydroperoxy)-3,5-dimethyl-1,2-dioxolane
Solvents: Acetonitrile; 100 min, 25 °C
- 1.2 **Reagents:** Sodium sulfite
Solvents: Water; 10 min, 25 °C

Experimental Protocols

31-488-CAS-7179296

Steps: 1 Yield: 64%

Dirhodium(II) Complexes of 2-(Sulfonylimino)pyrrolidine: Synthesis and Application in Catalytic Benzylic Oxidation

By: Wusiman, Abudurehman; et al

European Journal of Organic Chemistry (2012), 2012(16), 3088-3092, S3088/1-S3088/45.

- 1.1 **Catalysts:** Rhodium, tetrakis[μ-[*N*-(3,4-dihydro-2*H*-pyrrol-5-yl-κ*M*)methanesulfonamidato-κ*M*]di-, (*Rh-Rh*), stereoisomer
Solvents: Water; rt
- 1.2 **Reagents:** *tert*-Butyl hydroperoxide
Solvents: Water; 20 h, rt

Experimental Protocols

31-488-CAS-7310967

Steps: 1 Yield: 58%

Gold-catalyzed benzylic oxidation to carbonyl compounds

By: Li, Huanrong; et al

Tetrahedron (2009), 65(9), 1856-1858.

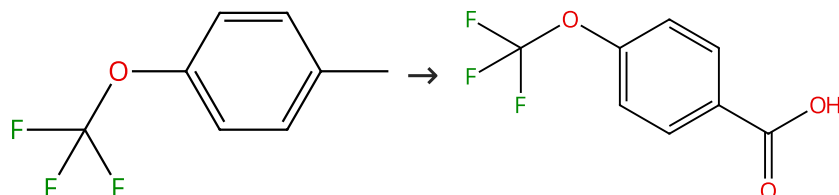
- 1.1 **Reagents:** *tert*-Butyl hydroperoxide, Pyridine, Oxygen
Catalysts: Potassium tetrachloroaurate
Solvents: Pyridine, Decane; rt; rt → 90 °C; 24 h, 90 °C; 90 °C → rt
- 1.2 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

31-488-CAS-12208043	Steps: 1 Yield: 53%	Iron-catalyzed benzylic oxidation with aqueous tert-butyl hydroperoxide
1.1 Reagents: <i>tert</i> -Butyl hydroperoxide Catalysts: Ferric chloride hexahydrate Solvents: Pyridine, Water; 24 h, 82 °C		By: Nakanishi, Masafumi; et al
1.2 Reagents: Hydrochloric acid Solvents: Water		Advanced Synthesis & Catalysis (2007), 349(6), 861-864.
Experimental Protocols		

Scheme 19 (1 Reaction)

Steps: 1 Yield: 86%



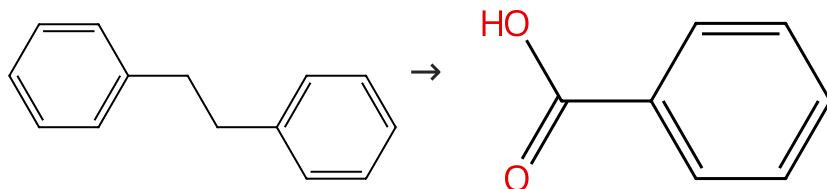
Suppliers (66)

Suppliers (87)

31-614-CAS-35375734	Steps: 1 Yield: 86%	Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds
1.1 Reagents: Oxygen Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1) Solvents: Acetonitrile; 18 h, 1 atm, 20 °C		By: Wu, Jieqing; et al
Experimental Protocols		Green Chemistry (2023), 25(3), 940-945.

Scheme 20 (2 Reactions)

Steps: 1 Yield: 46-86%



Suppliers (85)

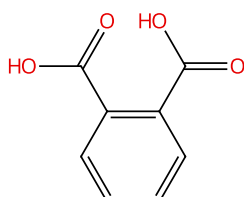
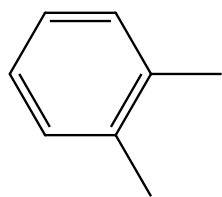
Suppliers (175)

31-614-CAS-35375749	Steps: 1 Yield: 86%	Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds
1.1 Reagents: Oxygen Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1) Solvents: Acetonitrile; 6 h, 1 atm, 20 °C		By: Wu, Jieqing; et al
Experimental Protocols		Green Chemistry (2023), 25(3), 940-945.

31-488-CAS-5454124	Steps: 1 Yield: 46%	Gold-catalyzed benzylic oxidation to carbonyl compounds
1.1 Reagents: <i>tert</i> -Butyl hydroperoxide, Pyridine, Oxygen Catalysts: Potassium tetrachloroaurate Solvents: Pyridine, Decane; rt; rt → 90 °C; 24 h, 90 °C; 90 °C → rt		By: Li, Huanrong; et al
1.2 Reagents: Hydrochloric acid Solvents: Water		Tetrahedron (2009), 65(9), 1856-1858.
Experimental Protocols		

Scheme 21 (1 Reaction)

Steps: 1 Yield: 84%



Suppliers (108)

Suppliers (117)

31-614-CAS-35375730

Steps: 1 Yield: 84%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 18 h, 1 atm, 20 °C

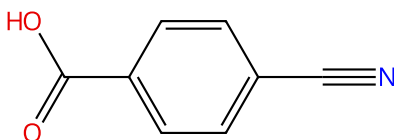
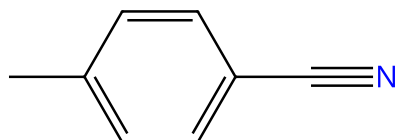
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 84%



Suppliers (73)

Suppliers (74)

31-614-CAS-35375715

Steps: 1 Yield: 84%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 18 h, 1 atm, 20 °C

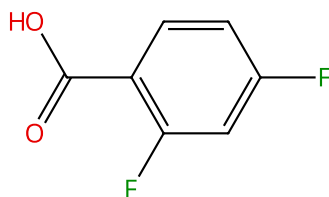
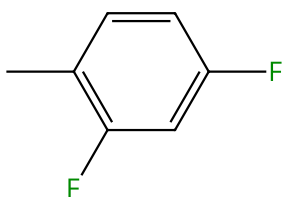
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 83%



Suppliers (62)

Suppliers (89)

31-614-CAS-35375722

Steps: 1 Yield: 83%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

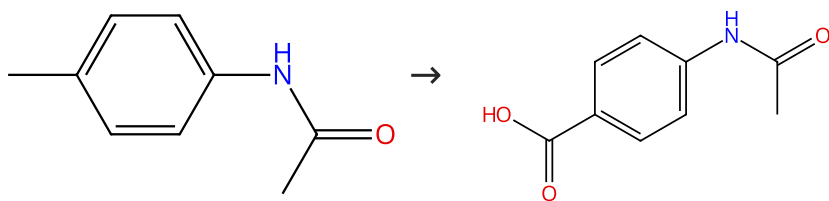
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 83%



Suppliers (75)

Suppliers (90)

31-614-CAS-35375723

Steps: 1 Yield: 83%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

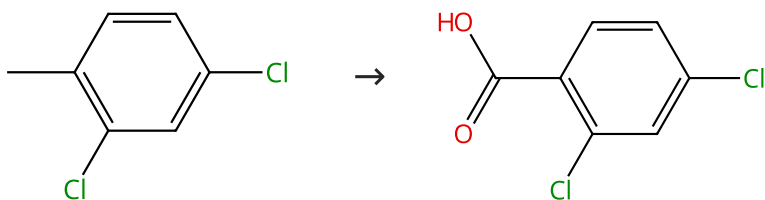
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 82%



Suppliers (56)

Suppliers (107)

31-614-CAS-35375724

Steps: 1 Yield: 82%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

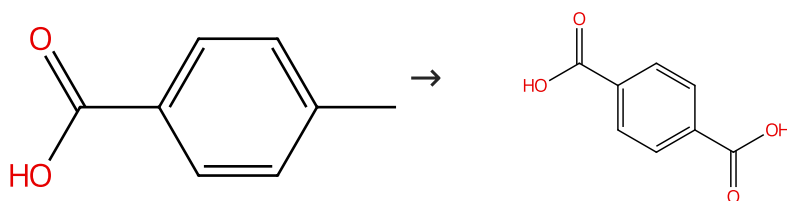
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 26 (1 Reaction)

Steps: 1 Yield: 82%



Suppliers (98)

Suppliers (106)

31-614-CAS-35375733

Steps: 1 Yield: 82%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 18 h, 1 atm, 20 °C

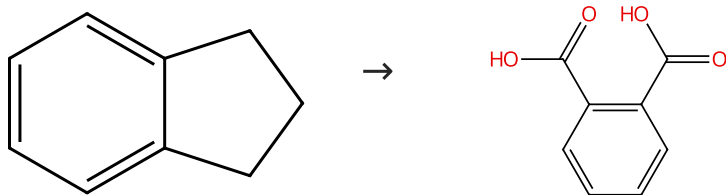
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 81%



Suppliers (90)

Suppliers (117)

31-614-CAS-35375743

Steps: 1 Yield: 81%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 18 h, 1 atm, 20 °C

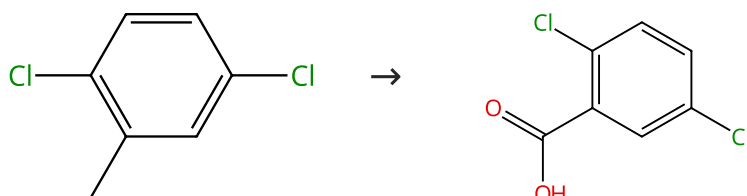
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 28 (1 Reaction)

Steps: 1 Yield: 80%



Suppliers (63)

Suppliers (82)

31-614-CAS-35375720

Steps: 1 Yield: 80%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

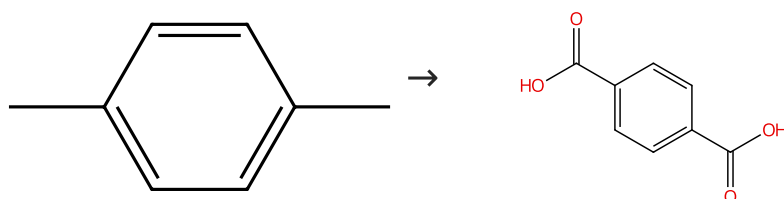
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 29 (1 Reaction)

Steps: 1 Yield: 80%



Suppliers (115)

Suppliers (106)

31-614-CAS-35375721

Steps: 1 Yield: 80%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 18 h, 1 atm, 20 °C

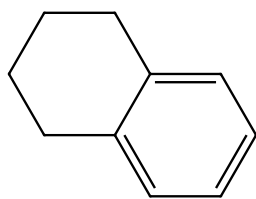
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

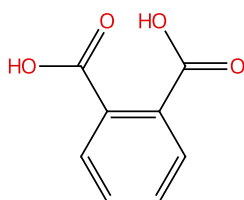
Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 78%



Suppliers (95)



Suppliers (117)

31-614-CAS-35375751

Steps: 1 Yield: 78%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 18 h, 1 atm, 20 °C

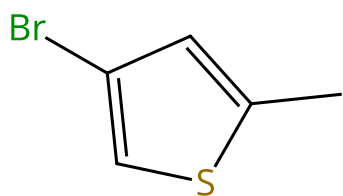
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

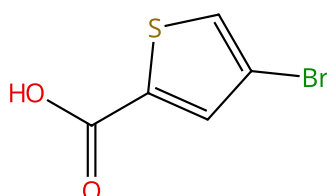
Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 74%



Suppliers (71)



Suppliers (99)

31-614-CAS-35375725

Steps: 1 Yield: 74%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

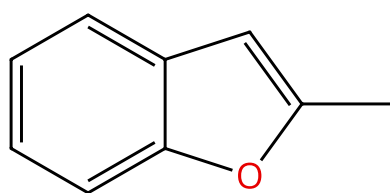
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

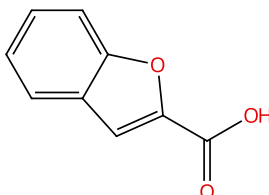
Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 74%



Suppliers (73)



Suppliers (99)

31-614-CAS-35375726

Steps: 1 Yield: 74%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

1.1 Reagents: Oxygen

Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)

Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

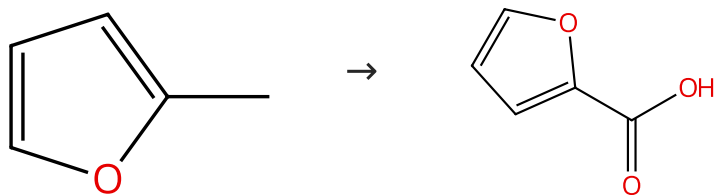
By: Wu, Jieqing; et al

Green Chemistry (2023), 25(3), 940-945.

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 57%



Suppliers (63)

Suppliers (96)

31-614-CAS-35375732

Steps: 1 Yield: 57%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

By: Wu, Jieqing; et al

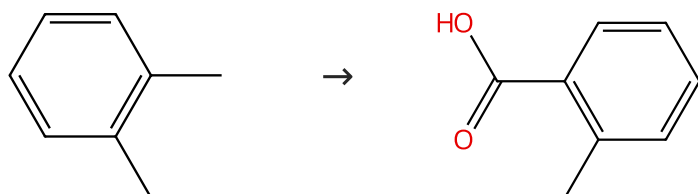
Green Chemistry (2023), 25(3), 940-945.

1.1 **Reagents:** Oxygen
Catalysts: Pyridinium, 1-(4-sulfobutyl)-, 1,1,1-trifluoro methanesulfonate (1:1)
Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

Experimental Protocols

Scheme 34 (1 Reaction)

Steps: 1 Yield: 55%



Suppliers (108)

Suppliers (92)

31-488-CAS-4081608

Steps: 1 Yield: 55%

Co(acac)₂-catalyzed allylic and benzylic oxidation by tert-butyl hydroperoxide

By: Han, Xiaoqiang; et al

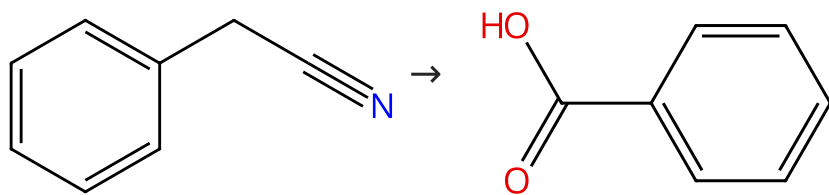
Synthesis (2013), 45(5), 615-620.

1.1 **Reagents:** *tert*-Butyl hydroperoxide
Catalysts: Cobalt(II) acetylacetonate
Solvents: Acetone; 28 °C; 15 h, 28 °C

Experimental Protocols

Scheme 35 (1 Reaction)

Steps: 1 Yield: 46%



Suppliers (48)

Suppliers (175)

31-488-CAS-16599714

Steps: 1 Yield: 46%

Iron-surfactant nanocomposite-catalyzed benzylic oxidation in water

By: Szabo, Fruzsina; et al

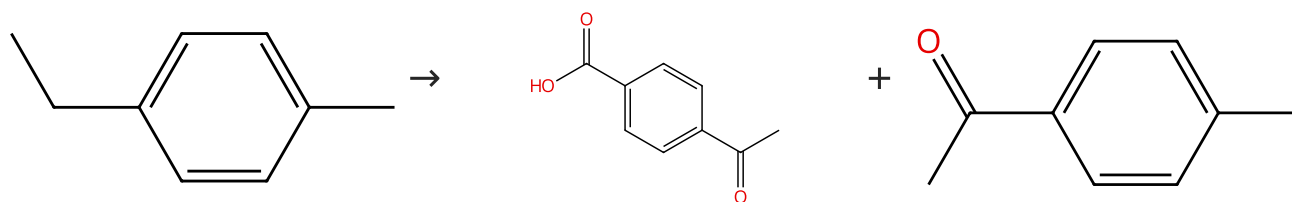
RSC Advances (2013), 3(15), 4903-4908.

1.1 **Reagents:** *tert*-Butyl hydroperoxide, Sodium dodecyl sulfate
Catalysts: Ferric sulfate
Solvents: Water; 24 h, 50 °C

Experimental Protocols

Scheme 36 (1 Reaction)

Steps: 1 Yield: 85%



Suppliers (77)

Suppliers (101)

Suppliers (103)

31-614-CAS-33766774

Steps: 1 Yield: 85%

Chemoselective Benzylic Oxidation of Alkyl-Substituted Aromatics to Ketones with trans-3,5-Dihydroperoxy-3,5-Dimethyl-1,2-Dioxolane

By: Najminejad, Z.; et al

Organic Preparations and Procedures International (2022), 54(6), 511-516.

1.1 **Reagents:** *rel*-(3*R*,5*R*)-3,5-Bis(hydroperoxy)-3,5-dimethyl-1,2-dioxolane

Solvents: Acetonitrile; 50 min, 25 °C

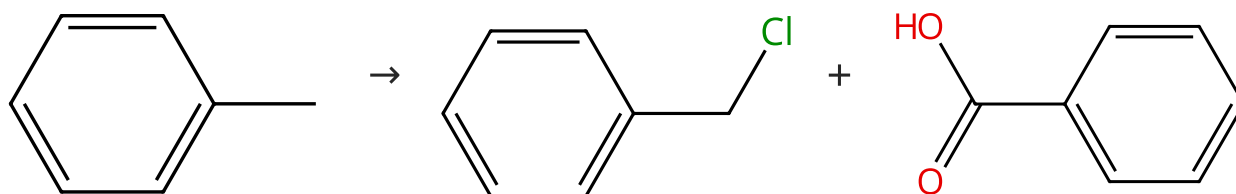
1.2 **Reagents:** Sodium sulfite

Solvents: Water; 10 min, 25 °C

Experimental Protocols

Scheme 37 (1 Reaction)

Steps: 1 Yield: 54%



Suppliers (280)

Suppliers (65)

Suppliers (175)

31-614-CAS-35375744

Steps: 1 Yield: 54%

Bronsted acid-catalysed aerobic photo-oxygenation of benzylic C-H bonds

By: Wu, Jieqing; et al

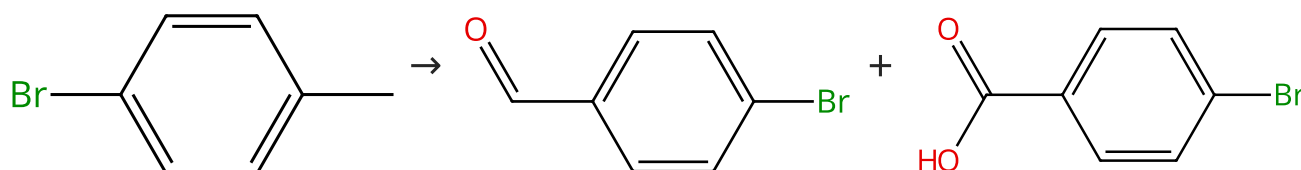
Green Chemistry (2023), 25(3), 940-945.

1.1 **Reagents:** Hydrochloric acid, Oxygen
Solvents: Acetonitrile; 6 h, 1 atm, 20 °C

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 41%



Suppliers (67)

Suppliers (90)

Suppliers (98)

31-488-CAS-9304506

Steps: 1 Yield: 41%

Dirhodium(II) Complexes of 2-(Sulfonylimino)pyrrolidine: Synthesis and Application in Catalytic Benzylic Oxidation

By: Wusiman, Abudurehman; et al

European Journal of Organic Chemistry (2012), 2012(16), 3088-3092, S3088/1-S3088/45.

1.1 **Catalysts:** Rhodium, tetrakis[μ-[*N*-(3,4-dihydro-2*H*-pyrrol-5-yl-κ*M*)methanesulfonamido-κ*M*]]di-, (*Rh*-*Rh*), stereoisomer
Solvents: Water; rt

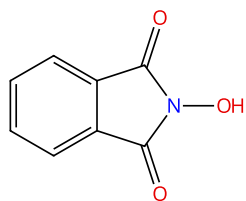
1.2 **Reagents:** *tert*-Butyl hydroperoxide

Solvents: Water; 20 h, rt

Experimental Protocols

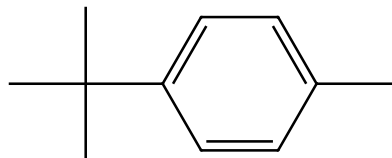
Scheme 39 (1 Reaction)

Steps: 1 Yield: 9%



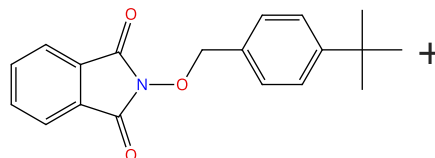
Suppliers (88)

+

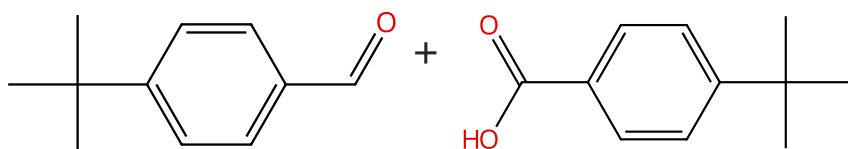


Suppliers (63)

→



Suppliers (2)



Suppliers (75)

Suppliers (92)

31-614-CAS-33763331

Steps: 1 Yield: 9%

1.1 **Reagents:** Pyridine, Borate(1-), tetrafluoro-, hydrogen, compd. with pyridine (1:1:1), Oxygen

Solvents: Acetonitrile, Dichloromethane; 40 - 60 min, rt

Experimental Protocols

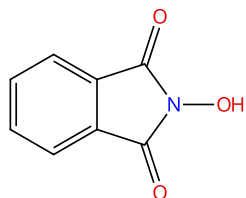
Electrochemical PINOylation of Methylarenes: Improving the Scope and Utility of Benzylic Oxidation through Mediated Electrolysis

By: Hoque, Asmaul Md; et al

Journal of the American Chemical Society (2022), 144(33), 15295-15302.

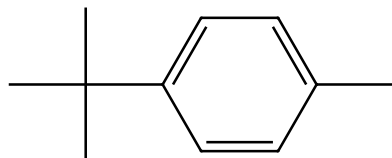
Scheme 40 (1 Reaction)

Steps: 1 Yield: 10%



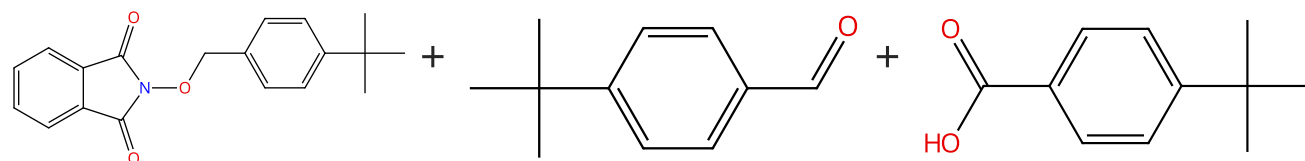
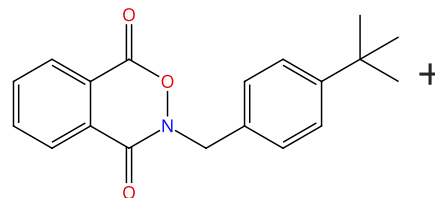
Suppliers (88)

+



Suppliers (63)

→



Suppliers (2)

Suppliers (75)

Suppliers (92)

31-614-CAS-33763334

Steps: 1 Yield: 10%

1.1 **Reagents:** Pyridine, Borate(1-), tetrafluoro-, hydrogen, compd. with pyridine (1:1:1), Oxygen

Solvents: Acetonitrile, Dichloromethane; 40 - 60 min, rt

Experimental Protocols

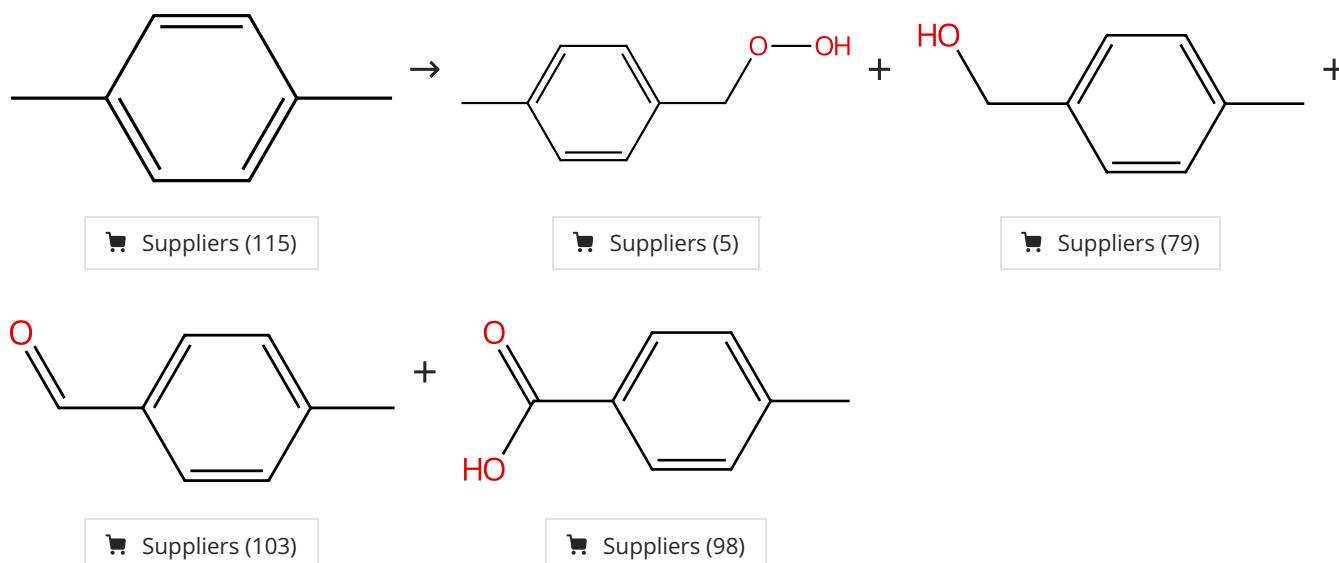
Electrochemical PINOylation of Methylarenes: Improving the Scope and Utility of Benzylic Oxidation through Mediated Electrolysis

By: Hoque, Asmaul Md; et al

Journal of the American Chemical Society (2022), 144(33), 15295-15302.

Scheme 41 (2 Reactions)

Steps: 1



31-207-CAS-15199607

Steps: 1

Benzylic oxidation and photooxidation by air in the presence of graphite and cyclohexene

By: Sereda, Grigoriy; et al

Tetrahedron Letters (2007), 48(19), 3417-3421.

1.1 Catalysts: Graphite

31-207-CAS-13070619

Steps: 1

Benzylic oxidation and photooxidation by air in the presence of graphite and cyclohexene

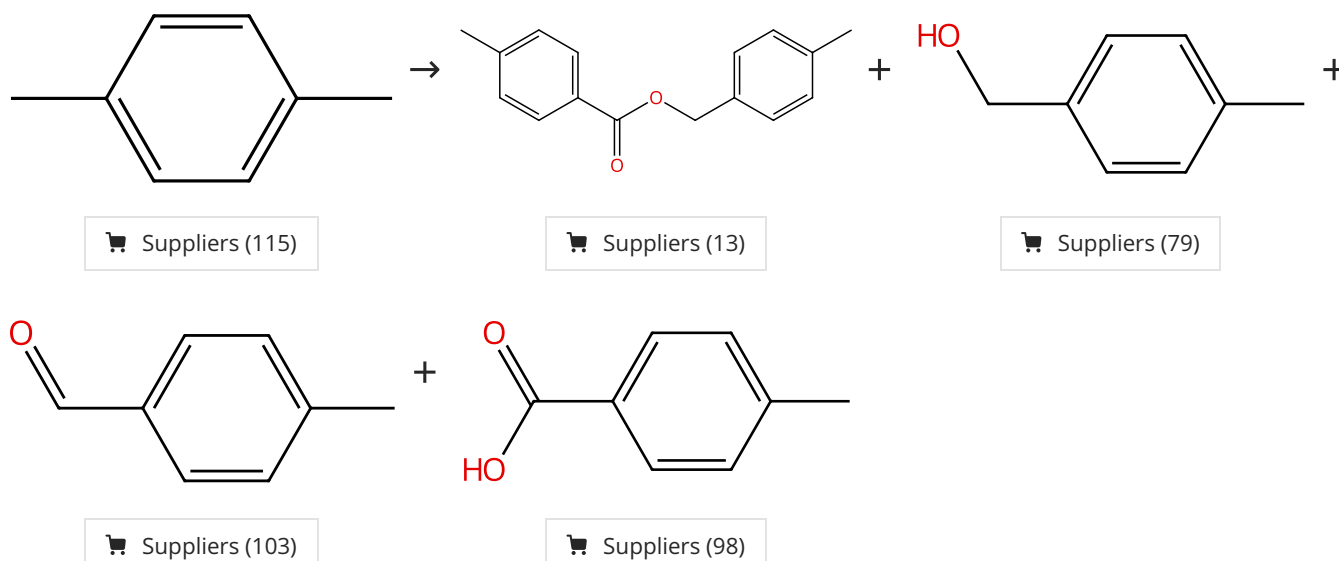
By: Sereda, Grigoriy; et al

Tetrahedron Letters (2007), 48(19), 3417-3421.

1.1 Reagents: Cyclohexene, Oxygen

Scheme 42 (1 Reaction)

Steps: 1



31-488-CAS-10933722

Steps: 1

Benzylic oxidation and photooxidation by air in the presence of graphite and cyclohexene

By: Sereda, Grigoriy; et al

Tetrahedron Letters (2007), 48(19), 3417-3421.

1.1 Reagents: Cyclohexene, Oxygen
Catalysts: Graphite

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